

(iii) MLE classifier is not affected by prior probability.

So: its output will be same as before (Sea bass)

But: Bayes Classifier (MAP classifier) considers the Prior probability. So: its decision may change.

a when $P(\text{Salmon}) = P(\text{Sea bass}) = \frac{1}{2}$

$$P(\text{salmon} | l=7) = \frac{\frac{1}{\sqrt{2\pi}} (0.135) * \frac{1}{2}}{P(l=7)} \quad [\text{see previous page}]$$

$$P(\text{sea bass} | l=7) = \frac{\frac{1}{\sqrt{2\pi}} (0.16) * \frac{1}{2}}{P(l=7)} \quad [\text{see previous page}]$$

b ~~So:~~
 $P(\text{Salmon}) = \frac{2}{3}$
 $P(\text{sea bass}) = \frac{1}{3}$

Longer. So Classify as Sea bass

$$P(\text{salmon} | l=7) = \frac{P(l=7 | \text{salmon}) * P(\text{salmon})}{P(l=7)} = \frac{\frac{1}{\sqrt{2\pi}} (0.135) * \frac{2}{3}}{P(l=7)}$$

$$P(\text{sea bass} | l=7) = \frac{\frac{1}{\sqrt{2\pi}} (0.16) * \frac{1}{3}}{P(l=7)}$$

$$= \frac{\frac{1}{\sqrt{2\pi}} * 0.09 * P(l=7)}{P(l=7)} = \frac{1}{\sqrt{2\pi}} * 0.053$$

Longer value
 So classify as Salmon

MATH 1.3

at Decision Boundary l , both $P(\text{Salmon})$ and $P(\text{Sea Bass})$ will be equal.

~~$P(l|\text{Salmon}) \neq P(l|\text{Sea Bass})$~~
For Bayes Classifier / MAP classifier, Start from Here:

$$P(\text{Salmon} | l) = P(\text{Sea Bass} | l)$$

$$\Rightarrow \frac{P(l | \text{Salmon}) * P(\text{Salmon})}{P(l)} = \frac{P(l | \text{Sea Bass}) * P(\text{Sea Bass})}{P(l)}$$

$\xrightarrow{\text{no prior given}}$ $\xrightarrow{\text{no prior given}}$

for MLE classifier, Just Start From Here:

$$\Rightarrow P(l | \text{Salmon}) = P(l | \text{Sea Bass})$$

$$\Rightarrow \frac{1}{\sigma_1 \sqrt{2\pi}} e^{-\frac{(l-\mu_1)^2}{2\sigma_1^2}} = \frac{1}{\sigma_2 \sqrt{2\pi}} e^{-\frac{(l-\mu_2)^2}{2\sigma_2^2}}$$

$$\Rightarrow \frac{1}{\sqrt{2\pi}} e^{-\frac{(l-5)^2}{2 \cdot 1}} = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(l-10)^2}{2 \cdot 4}}$$

$$\Rightarrow e^{-\frac{(l-5)^2}{2}} = \frac{1}{2} e^{-\frac{(l-10)^2}{8}}$$

take $\log_e$ of both sides

$$-\frac{(l-5)^2}{2} = -\log 2 - \frac{(l-10)^2}{8}$$

multiply by -8

$$4(l-5)^2 = 8 * (\log 2) + (l-10)^2$$

$$\Rightarrow 4l^2 - 40l + 100 = 5.54 + l^2 - 20l + 100$$

$$\Rightarrow 3l^2 - 20l = 5.54 \Rightarrow 3l^2 - 20l - 5.54 = 0$$

apply $ax^2+bx+c=0$

$$l = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Decision Boundary: 6.93 inch

no roots: $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Math 1.5

At decision Boundary l , we have $P(\text{Salmon} | l) = P(\text{Bass} | l)$

$$P(\text{Salmon} | \text{length} = l) = P(\text{Sea bass} | \text{length} = l)$$

$$\Rightarrow \frac{P(l | \text{Salmon}) * P(\text{Salmon})}{P(\text{length} = l)} = \frac{P(l | \text{Sea Bass}) * P(\text{Sea Bass})}{P(\text{length} = l)}$$

$$\Rightarrow P(l | \text{Salmon}) * \frac{2}{3} = P(l | \text{Sea Bass}) * \frac{1}{3}$$

$$\Rightarrow \frac{1}{1\sqrt{2\pi}} e^{-\frac{(l-5)^2}{2 \cdot 1}} * \frac{2}{3} = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(l-10)^2}{2 \cdot 4}} * \frac{1}{3}$$

$$\Rightarrow 2 e^{-\frac{(l-5)^2}{2}} = \frac{1}{2} e^{-\frac{(l-10)^2}{8}}$$

$$\Rightarrow 4 e^{-\frac{(l-5)^2}{2}} = e^{-\frac{(l-10)^2}{8}}$$

Take of Both sides $\log_e 4 - \frac{(l-5)^2}{2} = -\frac{(l-10)^2}{8}$

Multiply by 8 $8(\log_e 4) - 4(l-5)^2 = -(l-10)^2$
 $\Rightarrow 4(l-5)^2 - 8(\log_e 4) = (l-10)^2$

$$\Rightarrow 4l^2 - 40l + 100 - 11.09 = l^2 - 20l + 100$$

$$\Rightarrow 3l^2 - 20l - 11.09 = 0$$

$$0x^2 + bx + c = 0 \Rightarrow \text{roots} \leftarrow \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$l = \frac{7.18}{0.51}$$

Go: Decision Boundary $l = 14.1$

i) ML or MLE Classifier

$$\text{Likelihood of } l=7 = P(l=7 | \text{Salmon}) = \frac{1}{6\sqrt{2\pi}} e^{-\frac{(l-\mu)^2}{2\sigma^2}} = \frac{1}{6\sqrt{2\pi}} e^{-\frac{(7-5)^2}{2 \cdot 1}} = \frac{1}{6\sqrt{2\pi}} e^{-\frac{4}{2}} = \frac{1}{6\sqrt{2\pi}} e^{-2}$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{4}{2}} = \frac{1}{\sqrt{2\pi}} e^{-2} = \frac{1}{\sqrt{2\pi}} (0.135)$$

Likelihood of $l=7$

$$= P(l=7 | \text{sea bass}) = \frac{1}{6\sqrt{2\pi}} e^{-\frac{(l-\mu)^2}{2\sigma^2}} = \frac{1}{6\sqrt{2\pi}} e^{-\frac{(7-10)^2}{2 \cdot 4}} = \frac{1}{6\sqrt{2\pi}} e^{-\frac{9}{8}} = \frac{1}{6\sqrt{2\pi}} e^{-1.125} = \frac{1}{6\sqrt{2\pi}} (0.32)$$

$$= \frac{1}{\sqrt{2\pi}} (0.16)$$

Thus: $P(l=7 | \text{sea bass}) > P(l=7 | \text{Salmon})$

So: classify as Sea bass

ii) Bayes Classifier

$$P(\text{salmon} | l=7) = \frac{P(l=7 | \text{salmon}) * P(\text{salmon})}{P(l=7)}$$

as no prior probability given $= \frac{1}{2}$

$$P(\text{sea bass} | l=7) = \frac{P(l=7 | \text{sea bass}) * P(\text{sea bass})}{P(l=7)}$$

as no prior probability given $= \frac{1}{2}$

$$= \frac{\frac{1}{\sqrt{2\pi}} (0.135) * \frac{1}{2}}{P(l=7)}$$

$$= \frac{\frac{1}{\sqrt{2\pi}} (0.16) * \frac{1}{2}}{P(l=7)}$$

→ This one larger value

So: classify as Sea bass